Pull_OS - LEFT manual

Version: 2026-10-23. Hardware: diceFriend LEFT, Teensy 4.0.

Pull_OS is a granular pulse instrument built around emission, duration, formant motion, memory and phase-space feedback. In AS it behaves as one self-moving organism; in TS it becomes three separately pitched touch voices sharing the same granular character.

Main operation

- SHIFT selects the second control layer.
- Short DICE randomizes the active layer while preserving AS tempo.
- Hold unshifted DICE for about 3 seconds to switch AS/TS.
- Hold SHIFT + DICE for 10 seconds to enter the bootloader.
- Pot values use soft takeover after layer, mode and MIDI CC changes.

AS controls

Knob	NS layer	SHIFT layer
P1	Emission rate / fundamental	Grain overlap
P2	Grain duration	Memory length and feedback
P3	Formant position	Phaser motion
P4	Pulse shape	Wavefolding
P5	Chaos	Mutation depth
P6	Internal speed and release	Stereo space

AS pads act as performance gestures: T1 freezes memory, T2 ruptures the texture and T3 expands the phase space.

TS controls

Knob	NS layer	SHIFT layer
P1	Voice 1 pitch	Grain overlap
P2	Voice 2 pitch	Formant position
P3	Voice 3 pitch	Pulse shape / fold
P4	Voice 1 decay	Memory and phase space
P5	Voice 2 decay	Mutation
P6	Voice 3 decay	Stereo space

Each pad opens its corresponding voice. TS is externally played by pads, CV1 or MIDI; the internal AS clock does not trigger TS voices.

CV and USB MIDI

- CV1 is the priority trigger/clock input. In AS it advances the organism; in TS it rotates across the three voices.
- CV2 provides continuous modulation and sends CC74.
- Clock priority is CV, then MIDI, then internal. MIDI may take over after 20 seconds without CV; internal timing returns after 30 seconds without an external source.
- MIDI Clock uses 24 PPQN. Start, Continue and Stop control transport.
- In AS, Note On sets the emission pitch and triggers an event. Note Off releases the pitch override.
- In TS, Note On opens one of three voices. Note Off and Note On velocity 0 release the matching voice.
- CC20-31 control the twelve active panel values.

- Local steps and pads generate Note On/Off. Local clock and CV can generate Start and MIDI Clock.
- Incoming USB MIDI is not echoed back to the same port.

LEDs

The two layer LEDs show NS/SHIFT. The four lower LEDs display rungler motion, CV2 level and stereo audio activity.

First patch

Start in AS with P1 and P2 around one third, medium formant and low chaos. Raise overlap and memory gradually until individual impulses merge into a cloud. Use T1 to freeze a stable fragment, then touch T2 to break it apart.